Curriculum Vitae - Sebastian Michelmann

Academic positions

September 2023 onwards — **Assistant Professor (tenure track)**, *Department of Psychology, New York University*

June–August 2023 — **Associate Research Scholar**, *Princeton Neuroscience Institute, Princeton University*, Computational Memory Lab

2020–2023 (May) — **Postdoctoral Research Fellow**, *Princeton Neuroscience Institute, Princeton University*, Computational Memory Lab
German Research Foundation (DFG) funded project 437219953

2018–2020 — **Postdoctoral Research Associate**, *Princeton Neuroscience Institute, Princeton University*, Computational Memory Lab

2012–2014 — **Student Teaching Assistant**, *Department of Clinical Psychology, Biological Psychology and Psychotherapy Julius-Maximilians-Universität Würzburg*, Subjects: Introduction to Clinical Psychology (BSc), Clinical Psychology, Psychological Intervention and Clinical Neuropsychology (MSc)

2011–2012 — **Student Research Assistant**, *Department of Clinical Psychology, Biological Psychology and Psychotherapy, Julius-Maximilians-Universität Würzburg*, SSFB/Transregio 58 - TP B-1 Fear, Anxiety, Anxiety Disorders

Education

2018 (degree) — **PhD Psychology**, *School of Psychology, University of Birmingham, UK*, Memory and Oscillations Lab, Thesis title: "Temporal Dynamics and Mechanisms of Oscillatory Pattern Reinstatement in Human Episodic Memory."
2020 winner of the Glushko dissertation prize

2014 (degree) — **Diploma Psychology (equivalent to Master)**, *Julius-Maximilians-Universität Würzburg*, Würzburg, Germany, *GPA: 1.1*, non-compulsory subject: Computer Science
(GPA 1.1 equivalent to US/Canada: 3.93 out of 4)

2016–2018 — **Studies of Computer Science**, *Fern-Universität Hagen (distance learning)*, Hagen, Germany

2011–2014 — **Studies of Computer Science**, *Julius-Maximilians-Universität Würzburg*, Würzburg, Germany

2008–2014 — **Studies of Psychology**, *Julius-Maximilians-Universität Würzburg*, Würzburg, Germany

2010–2011 — **Studies of Psychology and Philology**, *Universidad de Salamanca*, Salamanca, Spain
Partnership program, additionally funded by PROMOS scholarship program of German Academic Exchange Service (DAAD)

2007 (degree) — **High school diploma**, *Bodelschwingh-Gymnasium Herchen*, Herchen, Germany, GPA: 1.3
(GPA 1.3 equivalent to US/Canada: 3.8 out of 4, 1^{st} percentile in cohort)

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Teaching

2025 (Spring) **Special Topics: Section 003, Episodic Memory**, *New York University*

2025 (Spring) **PSYCH-UA 25, Cognitive Neuroscience (Lecture)**, *New York University*

2024 (Fall) **PSYCH GA 3405 002, EEG/MEG/iEEG Methods (Seminar)**, *New York University*, Novel course created for graduates students in the program

2024 (Spring) **PSYCH-UA 25, Cognitive Neuroscience (Lecture)**, *New York University*

2019–2020 **NEU Fall Junior Tutorial 2019**, *Princeton Neuroscience Institute, Princeton University*

2012–2014 **Introduction to Clinical Psychology (BSc), and Clinical Psychology, Psychological Intervention and Clinical Neuropsychology (MSc)**, *Julius-Maximilians-Universität Würzburg*, Student Teaching Assistant

Professional memberships

2024 - ongoing **Memory Disorders Research Society (MDRS)**, *MDRS is a professional society dedicated to the study of memory. Members engage in basic and clinical research into how memory works and why it fails.*

2024 - 2025 **Society for Neuroscience (SFN)**, *World's largest organization of scientists and physicians devoted to advancing understanding of the brain and nervous system.*

2024 - 2025 **Association for Psychological Science (APS)**, *International non-profit organization with mission to promote, protect, and advance the interests of scientifically oriented psychology in research, application, teaching, and the improvement of human welfare.*

Funding – Awards

2022 **Post-Doctoral Fellow Award**, *Cognitive Neuroscience Society (CNS)*, CNS Annual Meeting, San Francisco

2020–2022 **Postdoctoral Research Fellowship**, *German Research Foundation (DFG)*, (project 437219953), funded for 24 months, extended for 6 months

2020 **Robert J. Glushko Prize for Outstanding Doctoral Dissertations in Cognitive Science**, *Cognitive Science Society*

2010 **PROMOS travel grant**, *German Academic Exchange Service (DAAD)*

2010–2011 **Scholarship of partnership with Universidad de Salamanca, Spain**, *Julius-Maximilians-Universität Würzburg, Germany*, the university annually selects one candidate per partner city and awards tuition, accommodation, and subsistence for the academic year

Service

September 2024 onwards **Center for Brain Imaging Steering Committee**, *Center for Brain Imaging, New York University*

2009–2010, 2011–2014 **Mentor for international students**, *International Office of Julius-Maximilians-Universität Würzburg*

Publications

Michelmann, **S.**, Kumar, M., Norman, K. A., & Toneva, M. (2025). Large language models can segment narrative events similarly to humans. *Behavior Research Methods*, *57*(1), 39.

Schmidt, B., Böhmer, J., Schnuerch, M., Koch, T., & **Michelmann**, **S.** (2024). Post-hypnotic suggestion improves confidence and speed of memory access with long-lasting effects. *Acta Psychologica*, *245*, 104240.

Zada, Z., Goldstein, A., **Michelmann**, **S.**, Simony, E., Price, A., Hasenfratz, L., Barham, E., Zadbood, A., Doyle, W., Friedman, D., Dugan, P., Melloni, L., Devore, S., Flinker, A., Devinsky, O., Nastase, S. A., & Hasson, U. (2024). A shared model-based linguistic space for transmitting our thoughts from brain to brain in natural conversations. *Neuron*, *112*(18), 3211–3222.e5.

Kumar, M., Goldstein, A., **Michelmann**, **S.**, Zacks, J. M., Hasson, U., & Norman, K. A. (2023). Bayesian surprise predicts human event segmentation in story listening. *Cognitive Science*, *47*(10), e13343.

Michelmann, **S.**, Hasson, U., & Norman, K. A. (2023). Evidence That Event Boundaries Are Access Points for Memory Retrieval. *Psychological Science*.

Rier, L., **Michelmann**, **S.**, Ritz, H., Shah, V., Hill, R. M., Osborne, J., Doyle, C., Holmes, N., Bowtell, R., Brookes, M. J., Norman, K. A., Hasson, U., Cohen, J. D., & Boto, E. (2023). Test-retest reliability of the human connectome: An OPM-MEG study. *Imaging Neuroscience*, *1*, 1–20.

D'Andrea, A., Basti, A., Tosoni, A., Guidotti, R., Chella, F., **Michelmann**, **S.**, Romani, G. L., Pizzella, V., & Marzetti, L. (2022). Magnetoencephalographic spectral fingerprints differentiate evidence accumulation from saccadic motor preparation in perceptual decision-making. *iScience*, 105246.

Wallace, G., Polcyn, S., Brooks, P. P., Mennen, A. C., Zhao, K., Scotti, P. S., **Michelmann**, **S.**, Li, K., Turk-Browne, N. B., Cohen, J. D., & Norman, K. A. (2022). RT-cloud: A cloud-based software framework to simplify and standardize real-time fMRI. *NeuroImage*, *257*, 119295.

Meconi, F., Linde-Domingo, J., S. Ferreira, C., **Michelmann**, **S.**, Staresina, B., Apperly, I. A., & Hanslmayr, S. (2021). EEG and fMRI evidence for autobiographical memory reactivation in empathy. *Human Brain Mapping*, *42*(14), 4448–4464.

Michelmann, **S.**, Price, A. R., Aubrey, B., Strauss, C. K., Doyle, W. K., Friedman, D., Dugan, P. C., Devinsky, O., Devore, S., Flinker, A., Hasson, U., & Norman, K. A. (2021). Moment-by-moment tracking of naturalistic learning and its underlying hippocampo-cortical interactions. *Nature Communications*, *12*(1), 5394.

Ngo, C. T., **Michelmann**, **S.**, Olson, I. R., & Newcombe, N. S. (2021). Pattern separation and pattern completion: Behaviorally separable processes? *Memory & Cognition*, *49*(1), 193–205.

Parish, G., **Michelmann**, **S.**, Hanslmayr, S., & Bowman, H. (2021). The Sync-Fire/deSync model: Modelling the reactivation of dynamic memories from cortical alpha oscillations. *Neuropsychologia*, *158*, 107867.

Treder, M. S., Charest, I., **Michelmann**, **S.**, Martín-Buro, M. C., Roux, F., Carceller-Benito, F., Ugalde-Canitrot, A., Rollings, D. T., Sawlani, V., Chelvarajah, R., Wimber, M., Hanslmayr, S., & Staresina, B. P. (2021). The hippocampus as the switchboard between perception and memory. *Proceedings of the National Academy of Sciences*, *118*(50), e2114171118.

Krzemiński, D., **Michelmann**, **S.**, Treder, M., & Santamaria, L. (2020). Classification of P300 Component Using a Riemannian Ensemble Approach. In J. Henriques, N. Neves, & P. de Carvalho (Eds.), *XV Mediterranean Conference on Medical and Biological Engineering and Computing – MEDICON 2019* (pp. 1885–1889). Springer International Publishing.

Griffiths, B. J., Parish, G., Roux, F., **Michelmann**, **S.**, Plas, M. v. d., Kolibius, L. D., Chelvarajah, R., Rollings, D. T., Sawlani, V., Hamer, H., Gollwitzer, S., Kreiselmeyer, G., Staresina, B., Wimber, M., & Hanslmayr, S. (2019). Directional coupling of slow and fast hippocampal gamma with neocortical alpha/beta oscillations in human episodic memory. *Proceedings of the National Academy of Sciences*, *116*(43), 21834–21842.

Michelmann, **S.**, Staresina, B. P., Bowman, H., & Hanslmayr, S. (2019). Speed of time-compressed forward replay flexibly changes in human episodic memory. *Nature Human Behaviour*, *3*(2), 143.

Michelmann, **S.**, Bowman, H., & Hanslmayr, S. (2018). Replay of stimulus-specific temporal patterns during associative memory formation. *Journal of Cognitive Neuroscience*, *30*(11), 1577–1589.

Michelmann, **S.**, Treder, M. S., Griffiths, B., Kerrén, C., Roux, F., Wimber, M., Rollings, D., Sawlani, V., Chelvarajah, R., Gollwitzer, S., Kreiselmeyer, G., Hamer, H., Bowman, H., Staresina, B., & Hanslmayr, S. (2018). Data-driven re-referencing of intracranial EEG based on independent component analysis (ICA). *Journal of Neuroscience Methods*, *307*, 125–137.

Andreatta, M., **Michelmann**, **S.**, Pauli, P., & Hewig, J. (2017). Learning processes underlying avoidance of negative outcomes. *Psychophysiology*, *54*(4).

Michelmann, **S.**, Bowman, H., & Hanslmayr, S. (2016). The temporal signature of memories: Identification of a general mechanism for dynamic memory replay in humans. *PLoS Biology*, *14*(8).

Staresina, B. P., **Michelmann**, **S.**, Bonnefond, M., Jensen, O., Axmacher, N., & Fell, J. (2016). Hippocampal pattern completion is linked to gamma power increases and alpha power decreases during recollection. *eLife*, *5*, e17397.

Book chapters

Parish, G. M., **Michelmann**, **S.**, & Hanslmayr, S. (2023). How should i re-reference my intracranial EEG data? In N. Axmacher (Ed.), *Intracranial EEG: A guide for cognitive neuroscientists* (pp. 451–473). Springer International Publishing.

Michelmann, **S.**, Griffiths, B. J., & Hanslmayr, S. (2022). The role of alpha and beta oscillations in the human EEG during perception and memory processes. In *The Oxford Handbook of EEG Frequency* (p. 202). Oxford University Press.

Preprints

Kerrén, C., **Michelmann**, **S.**, & Doeller, C. F. (2025, April 28). Hippocampal ripples initiate cortical dimensionality expansion for memory retrieval. https://doi.org/10.1101/2025.04.22.649929

Michelmann, **S.**, Dugan, P., Doyle, W., Friedman, D., Melloni, L., Strauss, C. K., Devore, S., Flinker, A., Devinsky, O., Hasson, U., & Norman, K. A. (2025, February 13). Fast-timescale hippocampal processes bridge between slowly unfurling neocortical states during memory search. https://doi.org/10.1101/2025.02.11.637471

Zada, Z., Nastase, S. A., Aubrey, B., Jalon, I., **Michelmann**, **S.**, Wang, H., Hasenfratz, L., Doyle, W., Friedman, D., Dugan, P., Melloni, L., Devore, S., Flinker, A., Devinsky, O., Goldstein, A., & Hasson, U. (2025, February 16). The “podcast” ECoG dataset for modeling neural activity during natural language comprehension. https://doi.org/10.1101/2025.02.14.638352

Pink, M., Vo, V. A., Wu, Q., Mu, J., Turek, J. S., Hasson, U., Norman, K. A., **Michelmann**, **S.**, Huth, A., & Toneva, M. (2024, October 10). Assessing episodic memory in LLMs with sequence order recall tasks. https://doi.org/10.48550/arXiv.2410.08133

Reviewing for peer reviewed journals

Nature Human Behaviour, Elife, PLOS Biology, Science Advances, Neuroscience & Biobehavioral Reviews, Scientific Reports, Psychological Review, Cerebral Cortex, Psychophysiology, Journal of Neuroscience Methods

Selected talks and conference contributions

Fast-timescale hippocampal processes bridge between slowly unfurling neocortical states during memory search (April 2025). Invited **talk** in "Cognitive Brown Bag (CBB)" talk series at Dartmouth University

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Bridging the gap between fast neural processes and long timescales in human episodic memory (January 2025). Invited **talk** in "Current Works in Behavior, Genetics, and Neuroscience" seminar series at Yale University

Michelmann S., Strauss, C., Doyle, W. K., Friedman, D., Melloni, L., Dugan, P., Devinsky, O., Devore, S., Flinker, A., Hasson, U., Norman, K. A. (October 2024) Tracing the neural underpinnings of memory search across slowly unfurling states. **Talk** in Nanosymposium presented at the *Annual Meeting of the Society for Neuroscience (SFN)*. Chicago, USA.

Michelmann S., Hasson, U., Norman, K. A. (May 2024) Evidence that event boundaries are access points for memory retrieval. **Talk** in Symposium presented at the *Association of Psychological Science – Annual Convention*. San Francisco, USA.

Michelmann S., Strauss, C., Doyle, W. K., Friedman, D., Melloni, L., Dugan, P., Devinsky, O., Devore, S., Flinker, A., Hasson, U., Norman, K. A. (April 2023): Tracking the reinstatement of event representations during continuous memory search with human ECoG. **Poster** presented at the *2023 International Conference on Learning and Memory*, Huntington Beach, USA

Michelmann S., Strauss, C., Doyle, W. K., Friedman, D., Dugan, P., Devinsky, O., Devore, S., Flinker, A., Hasson, U., Norman, K. A. (April 2022): Detecting slowly unfolding event patterns in naturalistic perception and memory with human ECoG. **Poster** presented at the *2022 Annual Meeting of the Cognitive Neuroscience Society (CNS)*, San Francisco, USA

Michelmann S., Price, A.R., Aubrey, B., Doyle, W.K. Friedman, D., Dugan, P.C., Devinsky, O., Devore, S., Flinker, A., Hasson, U., Norman, K.A. (August 2020): One shot learning of a naturalistic story improves predictions on a fast time-scale in the auditory cortex. **Talk** presented at the *2020 Context and Episodic Memory Symposium (CEMS)*, Virtual.

Michelmann, S., (August 2020) Temporal dynamics of episodic memory reinstatement. **Talk** at Glushko Dissertation Prize symposium, *2020 CogSci*, Virtual.

Michelmann, S., Staresina, B., S., Bowman, H., Hanslmayr S. (November, 2018) Forward and time-compressed replay of human episodic memories flexibly changes its speed. **Talk** in Nanosymposium presented at the *Annual Meeting of the Society for Neuroscience (SFN)*. San Diego, USA.

Michelmann, S., Bowman H., & Hanslmayr, S. (May, 2017). Searching for memory in brain oscillations - investigating the Synchronization/De-Synchronization Conundrum - **Talk** presented at *Memory Replay Workshop* at CUBRIC, Cardiff, UK

Michelmann, S., Bowman H., & Hanslmayr, S. (July, 2016). The temporal signature of dynamic memories. **Talk** presented at *ICOM-6 6th International Conference on Memory*, Budapest, Hungary.

Consulting

since 2020 **Cognition.run**, Consulting for the development of cognition.run, an online platform that facilitates behavioral research (www.cognition.run)

Languages

German (native), English (fluent), Spanish (fluent), French (proficient), Italian (proficient)

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